

NATIONAL BUREAU OF STATISTICS

Statistical Handbook

Every metric, every decision, and the reason for it

Nigeria Music Analytics System

Delivery 130 · 129 artists · Q1 2019 – Q3 2026

1 September 2026

NBS Statistical Handbook

Nigeria Music Analytics System — Delivery 130

Generated 01 September 2026. Every figure below is read from `04_Datasets` when this document is produced, so it cannot disagree with the data it describes.

This handbook contains no programming content. It is written for statisticians. Its purpose is that any number in this submission can be traced: what it is, where it came from, what was assumed, and why that decision was taken.

1. What this submission measures

The estimated revenue earned by Nigerian recording artists from digital music streaming platforms, quarter by quarter, from Q1 2019 to Q3 2026.

Artists	129
Quarters	31
Artist-quarter records	3,798
Gross streaming revenue, all quarters	\$469,833,610.14
In naira, at each quarter's own rate	₦502,390,063,070
Of which Q3 2026 (INCOMPLETE quarter)	\$28,424,707.67
Gross streaming revenue, 30 COMPLETE quarters	\$441,408,902.47

What it does not measure

This is important, and it is stated first rather than last.

- **It is not what platforms paid.** No platform reports payouts to this system. Revenue is estimated from audience figures using published per-unit rates.
- **It is not a count of streams.** No track-level stream count is available from any provider used here. Stream volumes are derived from audience counts.
- **It is not the whole Nigerian music industry.** It covers 129 artists, digital streaming only. Live performance, physical sales, sync, publishing, radio royalties and merchandising are outside it entirely.
- **It is not value added until the deduction in section 9 is applied.** Revenue is output, not GDP.

2. The population, and why it is 129 artists

The artists are those of the first submission to NBS. That list contains **131 rows** but describes **129 artists**, because 2 pairs of rows are the same person recorded twice under two different provider identities:

Kept	Merged into it	Why
Flavour	Flavour N'abania	One artist, two provider identities. Counting both inflates the artist count and charges the cost model for a person who does not exist.
Odunsi (The Engine)	Odunsi	One artist, two provider identities. Counting both inflates the artist count and charges the cost model for a person who does not exist.

Decision: count people, not records. A duplicate identity is merged into one artist, and the observations of the identity with the fuller record are used. **Reason:** an artist count is a count of economic units. Publishing a row count under the word "artists" would

overstate the population and every per-artist figure derived from it.

3. Residence, and the split between GDP and GNI

NBS asked: "Since the above-mentioned artists are living in those countries, their income will go to a different account which is not the production account. Kindly provide their revenue, cost of operation etc separate, it will be used to compile Gross National Income (GNI)."

This has been done. The classification is by **economic residence, not nationality**. An artist of Nigerian heritage who lives and works abroad is not part of Nigerian domestic production.

Classification	Artists	Gross revenue	Treatment
Resident in Nigeria	116	\$433,530,175.75	Domestic production account
Resident abroad, Nigerian heritage	12	\$36,303,414.38	Provided separately, for GNI compilation
Residence not established	1	\$20.08	NBS adjudication required

Files: Artist_Residency_Classification.csv (one row per artist, with the evidence for its classification), Domestic_Production_Account.csv , GNI_Diaspora_Account.csv .

A caution we place on the record. The diaspora figures are supplied *for use in* compiling GNI. They are not themselves GNI. What enters GNI is a specific income flow under the national accounts residency rules; that determination is NBS's, and this submission does not pre-empt it.

Basis of the classification. Residence is taken from the country the data provider registers against the artist. That is a provider's administrative record, not a residence determination. Every artist whose classification rests on it alone is flagged `needs_nbs_adjudication = Y`. **13 of 129 artists carry that flag.** NBS should treat those as candidates to confirm, not as settled.

4. Revenue by product

NBS asked: "Revenue generated and operating cost is broken-down according to product, example Spotify, Youtube, Deezer and others to enable us know the driver of digital music among them products."

File: Revenue_And_Cost_By_Platform.csv — one row per platform, per quarter, per account, with revenue, allocated cost and net result.

Product	Gross revenue, all quarters	Share	What the figure rests on
Spotify	\$238,655,190.81	50.8%	Observed monthly listeners
YouTube	\$154,288,857.96	32.8%	Channel view volume
Deezer	\$5,293,004.03	1.1%	Observed fan counts
Other digital platforms	\$71,596,557.34	15.2%	Nothing observed — an assumption

The driver of digital music revenue in this dataset is Spotify, followed by YouTube. Deezer is small. The fourth line is not a measurement and must not be read as one: it is a flat uplift applied to Spotify revenue to stand for platforms that were never queried (Apple Music, Amazon, Boomplay, Audiomack, Tidal). No figure from any of those platforms appears anywhere in this submission.

5. How each revenue figure is calculated

5.1 Spotify — and the question NBS raised

NBS asked: "Kindly give adequate explanation on what you mean by Spotify streams monthly listeners $\times$ 3.5 streams/listener/month $\times$ 3 months. Are you saying that the number of streams/listeners are constant?"

The question was correct, and the method has been changed.

The first submission took ONE listener figure for a quarter and multiplied it by three months. That does assume the listener count is constant across the quarter. It is not: it moves every month.

The quarter is now built from **each month's own observed listener level**:

Quarterly streams = (Month 1 listeners $\times$ 3.5) + (Month 2 listeners $\times$ 3.5) + (Month 3 listeners $\times$ 3.5)

Equivalently, the mean of the observed monthly levels carried across three months. Where all three months are observed the two are arithmetically identical.

Coverage of that change, across all 3,798 records:

Monthly levels observed in the quarter	Records	Share
3 months	3,397	89.4%
2 months	235	6.2%
1 month	15	0.4%
no monthly level	151	4.0%

Every record states its own basis in the column `spotify_listeners_basis`.

What remains an assumption. The figure of **3.5 streams per listener per month** is not measured by this system and is not reported by Spotify. It is an industry proxy. Revenue scales linearly with it: if the true figure is 7, Spotify revenue doubles. It is the single most consequential assumption in this submission, and NBS may substitute its own value — the effect is proportional and easy to restate.

What is measured. The monthly listener counts themselves are observed from the provider. What is assumed is how many times each listener plays a track.

5.2 YouTube

Revenue = quarterly view volume $\times$ \$0.004 per view.

View volume is a *cumulative counter*: the platform reports total views to date, so the quarter's volume is the increase across the quarter, not the level.

Basis of the view volume	Records	Share
Observed increase in the counter	1,876	49.4%
Estimated (no usable observation)	919	24.2%
No YouTube presence found	1,003	26.4%

Two decisions worth stating.

Counter restatements are removed. A cumulative counter is sometimes restated by the provider — channels merge, history is back-filled — and it leaps in one day by more than the channel earns in years. One artist gained 826 million views on a single day against a normal day of 351,682. Those are bookkeeping events, not Nigerians watching videos, and counting them as revenue would be wrong. Any day moving more than 50 times the artist's own typical daily rate is repriced at that typical rate.

A quarter must be spanned to be measured. If the observations cover only part of the quarter, the increase across them is not the quarter's volume. Where coverage falls below 90% the observation is treated as inadequate and an estimate is used instead — and the record says so. Q3 2021 was measured over 8 days of 92 and published a fall of 11.6% that never happened.

5.3 Deezer

Revenue = fans × 2.0 streams per fan per month × 3 months × \$0.004. Fan counts are observed; the 2.0 is an assumption of the same kind as the Spotify 3.5.

5.4 Other digital platforms

A flat **30% uplift on Spotify revenue**, standing for platforms never queried. It is an assumption end to end and is carried in its own column, outside the platform breakdown, so it cannot be mistaken for a measured platform.

6. Costs, and how they are attributed to products

NBS said: *"The break down is commendable."* — the four categories are kept unchanged.

Category	Basis
Studio Production	₦750,000 per artist per quarter (ASM)
Digital Distribution	₦15,000 per artist per quarter (ASM)
Promotion & Marketing	₦250,000 per artist per quarter (ASM)
Web Hosting & CDN	₦50,000 per artist per quarter (ASM)

On attributing cost to a product, we must be straightforward with NBS.

NBS asked for operating cost broken down by product. It is provided, but with an important qualification: **no cost in this dataset is directly attributable to a platform.** The cost card is measured per artist per quarter — a studio session, a distribution fee, a promotion budget — none of which arrives labelled with the platform it served.

Every cost is therefore a **shared cost**, allocated to platforms **on each platform's share of revenue in that quarter and account.** The allocation basis is printed on every row of `Revenue_And_Cost_By_Platform.csv`, in the column `cost_allocation_basis`, and the direct-cost column is zero throughout — because it is genuinely zero, not because it was not computed.

Why revenue share. It is the conventional allocator where no usage measure exists, and it is transparent. NBS may prefer another basis; the revenue shares are published beside the allocation so any alternative can be applied.

What the unit costs are. They are assumptions drawn from secondary sources, not observations of these artists' spending. Only the **artist counts** they are multiplied by are measured.

7. Currency conversion

Every platform pays in US dollars. The naira figures are a **conversion**, not a measurement, and the rate chosen decides the answer.

A correction we must report. The first submission applied a single flat rate of ₦1,500 per dollar to every quarter from 2019 to 2026. The naira did not sit still: it moved from about ₦307 to about ₦1,530. Applying ₦1,500 to 2019 overstated the 2019 figure roughly fivefold.

This matters more here than almost anywhere else, because NBS is rebasing GDP onto a **2019 base year**. A base-year figure inflated fivefold by a conversion rate would carry through the whole rebased series.

Year	Rate applied	2019 comparison
2019	306.92	the base year
2020	358.81	
2021	401.15	
2022	425.98	
2023	645.00	
2024	1478.00	
2025	1530.00	
2026	1530.00	

Rates are annual averages of the official rate, applied to each quarter of their year. They are an **assumption**, held separately from the revenue logic precisely so **NBS can replace them wholesale** with its own official series. Substituting a different table changes only the naira columns; the dollar measurement is untouched.

Known simplifications: annual averages rather than quarterly, so within-year movement is not captured; one rate for all flows, with no distinction between official, NAFEM and parallel rates; and 2026 carries the 2025 rate because 2026 is not a complete year. Each is a judgement NBS may wish to override.

8. The back-cast to 2019

NBS said: "before the digital music data will be incorporated to GDP series, it has to be back casted down to 2019 new base year for it to align with the series otherwise it will give very high."

The series runs from **Q1 2019**, the new base year, to Q3 2026 — 31 quarters.

Year	Gross revenue (USD)	Gross revenue (NGN, period rate)
2019	\$11,358,001.91	₦3,485,997,946
2020	\$19,022,130.49	₦6,825,330,641
2021	\$32,540,082.70	₦13,053,454,175
2022	\$58,603,839.77	₦24,964,063,665
2023	\$83,815,499.76	₦54,060,997,345
2024	\$89,917,031.40	₦132,897,372,409
2025	\$93,887,636.69	₦143,648,084,136
2026	\$80,689,387.42	₦123,454,762,753

How the early years were built. The providers' own history does not reach back uniformly to 2019. Where a series begins later than 2019, the earlier quarters are estimated by a stated rule and every such record is labelled:

- **Spotify listeners** begin 2019-05-16. Q1 2019 levels are carried back from each artist's own first observed quarter, deflated by the growth the data itself shows between the first two observed quarters. Records carry `spotify_listeners_source = estimated`.
- **YouTube view history** begins 2021-09-23. For quarters before that, view volume is estimated from subscriber counts at a rate calibrated on the observed period and extrapolated backwards. Records carry `youtube_views_source = estimated`.

This is the weakest part of the submission and we say so plainly. No YouTube view observation exists anywhere before September 2021, so YouTube revenue for 2019 to mid-2021 rests entirely on a model. It is labelled on every record, and a reader can remove it by filtering on that column.

9. National accounts treatment

Revenue is not value added, and this submission does not present it as such.

File: National_Accounts_Aggregates.csv

Concept	Treatment here
Output	Gross streaming revenue
Intermediate consumption	The four cost categories, all purchased services
Gross value added	Output less intermediate consumption
Compensation of employees	NOT MEASURED. The cost card has no labour component
Operating surplus / mixed income	The whole of value added, by consequence

This treatment requires NBS confirmation. Whether each cost category is intermediate consumption under SNA, and whether any part of the artists' own return is compensation of employees rather than mixed income, are national-accounts judgements. We have set out the components so NBS can classify them; we have not assumed the answer, and compensation of employees is left blank rather than zero because it is unmeasured, not nil.

10. Every assumption, in one table

Nothing below is measured. Each is a number chosen for a stated reason, and each can be replaced.

Constant	Value	Class	What it does	Source	Limitation
STREAMS_PER_LISTENER_MONTH	3.5	EST	Converts Spotify monthly listeners (reach) into plays (volume).	Industry proxy. Not derived from Nigerian data.	Held constant for every artist, quarter and year. Revenue level scales linearly with it: at 5.0 revenue would be 43% higher, at 2.0 43% lower. Cannot be replaced without track-level stream counts, which both providers deny (HTTP 401).
SPOTIFY_PER_STREAM	0.004	EST	Payout per Spotify stream.	Industry average (Ditto Music 2026, Chartlex 2026).	Flat across all 31 quarters. No Nigerian rate card obtained; actual payouts vary by territory, subscription tier and distributor.
YOUTUBE_PER_VIEW	0.004	EST	Payout per YouTube view.	Industry average (Hootsuite 2025).	Flat across all quarters; real RPM varies by territory and format.
DEEZER_PER_STREAM	0.004	EST	Payout per Deezer stream.	Industry average.	Deezer is under 1% of total revenue, so sensitivity is negligible.
DEEZER_STREAMS_PER_FAN_MONTH	2.0	EST	Converts Deezer fans into plays.	Industry proxy.	Same structural weakness as the Spotify multiplier.

Constant	Value	Class	What it does	Source	Limitation
VIEWS_PER_SUBSCRIBER_MONTH	13.527	EST	Estimates YouTube views where no observed quarter volume exists (all quarters before Q3 2021, quarters whose observations do not span the period, and artists the views series never covers).	Calibrated: OLS on the AGGREGATE observed ratio across the 19 fully-observed quarters Q4 2021 - Q2 2026, $R^2 = 0.782$. Replaces the first submission's unsourced flat 15.0.	The fallback era lies BEFORE the observed window, so the rate there is an extrapolation, not a measurement; it is floored at the lowest observed ratio (5.99). Applied ONLY where observation is absent or inadequate; every row carries youtube_views_source.
COUNTER_RESTATEMENT_FACTOR	50.0	ASM	Above this, a one-interval jump in a cumulative counter is a provider restatement (channel merge or backfill), not consumption, and is repriced at the median rate.	Set from the observed distribution: 44 of 2,168 artist-quarters exceed 50x, headed by a single day of 826,001,508 views against a 351,682 median day. No genuine quarter approaches it.	A judgement threshold. It cannot distinguish a restatement from a genuine viral event of the same size; it is set far above any observed organic day so that trade-off never binds in practice.
VIEWS_PER_LISTENER_MONTH	12.48	EST	Estimates YouTube views for artists holding a YouTube listener level but no observed view volume and no subscriber level.	Calibrated: $\text{aggregate sum}(\text{views})/\text{sum}(\text{listeners})/3$ over the 1,703 artist-quarters carrying both series.	A listener is a monthly-audience figure, not a follower; the ratio is measured on artists who have both series and may not transfer to those who have only one. Labelled per row.

Constant	Value	Class	What it does	Source	Limitation
MIN_OBSERVED_SPAN_COVERAGE	0.9	ASM	Below this, a cumulative counter's (last - first) delta measures a shorter window than the quarter and is not a valid quarterly volume, so the modelled fallback is used instead.	Set from observed coverage: 19 quarters span 97.8-100%, Q3 2021 spans 8.7% and Q3 2026 (unfinished) 42.4%. The threshold separates those two from every adequately observed quarter.	A judgement threshold, not a measurement. Quarters it rejects are labelled estimated rather than silently published low.
UNMEASURED_UPLIFT_RATE	0.3	ASM	Uplift for platforms never queried (Apple Music, Amazon, Boomplay, Audiomack and others).	Assumed from Spotify's approximate market share. Verified against the delivered file, which reproduces at exactly this value.	NOT a platform and must never be presented as one. No Boomplay, Audiomack, Apple Music or Amazon revenue is measured anywhere in it.
NAIRA_PER_USD	1500	ASM	Fixed conversion for all naira figures.	Single assumed rate.	The real NGN/USD rate moved materially across 2019-2026. Every naira figure in the delivery is therefore a constant-rate conversion, not a market conversion, and cross-year naira comparisons are affected.
TRACKS_PER_QUARTER	2	ASM	Multiplier for per-track cost categories.	Assumption.	REPLACEABLE: actual release dates for 100,019 songs are now held in Artist_Catalogue_Summary.csv and could replace this with a counted value.
AVG_PRODUCTION_COST_NGN	750000	ASM	Studio production cost.	NigerianInformer 2025 (secondary).	No measured cost input exists anywhere in the pipeline.
AVG_DISTRIBUTION_COST_NGN	15000	ASM	Digital distribution cost.	Bliss hype 2026 (secondary).	No measured cost input.
AVG_PROMOTION_COST_NGN	250000	ASM	Promotion and marketing cost.	TaGetMedia 2025 (secondary).	No measured cost input.
AVG_HOSTING_COST_QUARTERLY_NGN	50000	ASM	Web hosting and CDN cost.	Industry estimate (secondary).	No measured cost input.

EST — estimated from observed data by a stated rule. **ASM** — assumed from a source outside this system. Neither is an observation and neither is presented as one.

11. How to read a value's status

Every value in this submission carries one of these:

Code	Meaning
OBS	Observed directly from a provider
CNT	Counted from observed records
AGG	Aggregated from observations
EST	Estimated from observed data by a stated rule
ASM	Assumed from a source outside this system
UNK	Not measured. Never rendered as zero

The distinction between UNK and zero is the most important convention in this submission. A blank means the quantity was not measured. A zero means it was measured and found to be nil. They are never interchangeable. Where the domestic/export split could not be measured — every quarter before Q1 2021 — the cells are blank, not zero, because publishing zero would assert that Nigerian artists earned nothing abroad in those years.

Measure	Quarters measured	Quarters blank
Domestic / export split	23	8

12. Checks NBS can run

Every published figure can be recomputed from other published columns. These hold on all 3,798 records and can be tested in a spreadsheet:

Check	Identity
Spotify revenue	$\text{listeners} \times 3.5 \times 3 \times 0.004$
YouTube revenue	$\text{views} \times 0.004$
Deezer revenue	$\text{fans} \times 2.0 \times 3 \times 0.004$
Other platforms	$\text{Spotify revenue} \times 0.30$
Gross revenue	the four above, summed
Naira	$\text{gross revenue} \times \text{that quarter's rate}$
Quarter total	the sum of that quarter's artist records
Account split	$\text{domestic} + \text{diaspora} + \text{unclassified} = \text{total}$

One warning. `Gross_Streaming_Revenue.csv` and `Gross_Export_Revenue.csv` each contain a `=== PERIOD TOTAL ===` row per quarter, reproduced because the first submission carried them. **Exclude those rows when summing.** Including them doubles every total exactly.

13. What changed since the first submission, and why

Change	Reason
Series extended to 31 quarters from Q1 2019	NBS requires the 2019 base year
Naira converted at each year's own rate	A flat ₦1,500 overstated 2019 by ~5×

Change	Reason
Spotify built from each month's own level	NBS asked whether listeners were constant. They were being treated as constant
Diaspora artists separated	NBS requires them outside the production account
Revenue and cost broken down by platform	NBS asked, to identify the driver
Output, intermediate consumption and GVA separated	Revenue is not value added
Two duplicate artists merged	One person counted twice inflates the population and the cost model
Export markets computed per artist	The first submission wrote one identical market list on every row; it was not a detection
Provider counter restatements removed	A single day of 826 million views is a bookkeeping event, not consumption
Every record states its own basis	So an estimate can never be read as an observation

14. Limitations, stated plainly

1. **No platform payout is observed.** All revenue is estimated from audience.
2. **No track-level stream count exists** in any source used here.
3. **The 3.5 and 2.0 multipliers are proxies.** Revenue scales linearly with them.
4. **The 30% other-platform uplift measures nothing.** No such platform was queried.
5. **YouTube before September 2021 is entirely modelled.** No view observation exists.
6. **Residence rests on a provider's administrative country field** for most artists.
7. **Exchange rates are annual averages** of one rate, not quarterly and not differentiated by market.
8. **Q3 2026 is incomplete** — observations end 11 August 2026, 42 of 92 days — yet its level-based components bill three full months. It is 6.05% of the headline. The 30-complete-quarter total is published beside it for this reason.
9. **Employment is national**, from secondary sources. It is not these artists' employees and is not scaled to them.

15. Authorship

Authorship and attribution remain subject to NBS review and approval, following completion of the full back-cast dataset and presentation to the Statistician General of the Federation. Nothing in this submission should be read as claiming that authorship has been settled.

16. Glossary

Term	Meaning in this submission
Monthly listeners	Distinct accounts that played an artist in the preceding 30 days, as reported by the platform. A <i>level</i> , not a count of plays
Stream	One play of a track. Not observed here; derived from listeners
Channel views	Cumulative total views of an artist's YouTube channel. The quarter's volume is the increase across the quarter
Subscribers / fans	Standing followers of a channel or profile. A level
Gross streaming revenue	Estimated artist-side revenue before any cost
Domestic / export split	The share of listening inside and outside Nigeria, from observed listener geography
Back-cast	A figure for a period earlier than the provider's own history, produced by a stated rule
Economic residence	Where an artist lives and works. Distinct from nationality

Term	Meaning in this submission
Period total row	A summary row inside the data file. Exclude when summing